

Intel and AMD unite!

Intel and AMD – archrivals for decades – are teaming up. Yes, you read that right – I nearly did a double-take myself. The two giants of the computing world have announced the formation of an x86 ecosystem advisory group. Their common goal? To accelerate innovation and keep the x86 platform not just alive, but thriving in an era where the competition is fiercer than ever.

The announcement of the x86 ecosystem advisory group, which includes tech luminaries like Linus Torvalds and Tim Sweeney, along with founding members such as Broadcom, Dell, Google, HPE, HP Inc., Lenovo, Meta, Microsoft, Oracle, and Red Hat, signals a significant shift in strategy. “We are on the cusp of one of the most significant shifts in the x86 architecture and ecosystem in decades,” said Pat Gelsinger, Intel’s CEO. Lisa Su, AMD’s Chair and CEO, echoed this sentiment: “Establishing the x86 Ecosystem Advisory Group will ensure that the x86 architecture continues evolving as the compute platform of choice for both developers and customers.”

So, why now? Why are Intel and AMD setting aside their fierce rivalry to collaborate? The answer lies in the meteoric rise of ARM-based architectures. Apple’s introduction of its M1 and subsequent M-series chips has been nothing short of revolutionary. These chips deliver impressive performance with remarkable energy efficiency, challenging the long-held belief that x86 processors are the only viable option for high-performance computing tasks.

Qualcomm and MediaTek are also pushing ARM-based processors into new territories, with ambitions to power everything from laptops to data centers. Even Microsoft has been experimenting with ARM-based Windows machines, and let’s not forget that AWS offers ARM-based instances in its cloud services.

Let’s not forget NVIDIA, the behemoth of graphics processing that’s been making significant inroads into the CPU arena, especially with its sights set on AI and machine learning. While Intel and AMD have dominated traditional computing with x86 architectures, NVIDIA’s focus on ARM-based processors for AI workloads presents a formidable challenge. ARM’s architecture offers efficiency and scalability that



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x86 has seemingly struggled to match in the rapidly evolving AI landscape. The emergence of AI has shifted the computing battleground. ARM architectures, with their power efficiency and flexibility, have become the go-to for AI applications, AI accelerated workloads, and even high-performance computing. NVIDIA’s advancements in AI accelerators and their ARM-based initiatives are eating into territories once securely held by x86 veterans like Intel and AMD. The ARM ecosystem is growing, and it’s not just about hardware. The software support for ARM architectures has improved dramatically, making it easier for developers to build and optimise applications for these platforms.

Intel and AMD’s creation of the x86 ecosystem advisory group is a strategic move to counter this computing shift to ARM-based platforms. By collaborating, Intel and AMD no doubt aim to streamline development, ensure compatibility, and rejuvenate the x86 platform. Their common goal is to make it more appealing for developers to create innovative solutions without the fragmentation that has sometimes plagued the x86 ecosystem. The advisory group aims to “unite industry leaders to shape the future of x86 and foster developer innovation through a more unified set of instructions and architectural interfaces.” This means simplifying architectural guidelines to enhance software consistency and interfaces across x86 products from both Intel and AMD. It’s too early to tell whether this alliance will reinvigorate the x86 platform or merely slow the tide of ARM’s advance. The computing landscape is evolving rapidly, with AI workloads, custom chiplets, and advancements in 3D packaging reshaping what’s possible.

The success of this initiative will likely hinge on whether Intel and AMD can genuinely collaborate to deliver meaningful improvements without stifling competition. Execution will be key. As someone who’s witnessed the ebb and flow of tech battles over the years, I can’t help but be intrigued. The merging of forces between Intel and AMD could lead to innovations we haven’t even imagined yet. Or it could be too little, too late. Either way, the computing wars are heating up once again. **d**